5.10.4 Construct a quadrilateral when its 3 sides and 2 angles are given

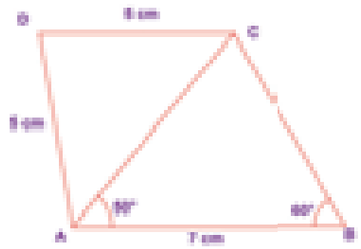
Example 5.24

Construct a quadrilateral $ABCD$ with AB=7 cm, AD= 5 cm, CD = 5 cm, $\angle BAC = 50°$ and $\angle ABC = 60°$. Also find its area.

Solution:

Given:

AB=7 cm, AD= 5 cm, CD = 5 cm and two angles $\angle BAC = 50°$ and $\angle ABC = 60°$



Rough diagram

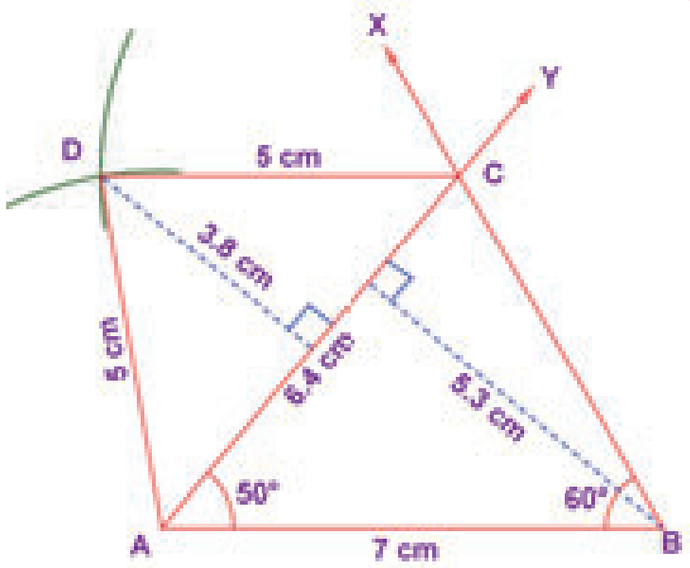


Fig. 5.43

Steps:

1. Draw a line segment AB = 7 cm.
2. At A on AB, make $\angle BAY = 50°$ and at B on AB, make $\angle ABX = 60°$. Let them intersect at C.
3. With A and C as centres, draw arcs of radius 5 cm each. Let them intersect at D.
4. Join AD and CD.
5. $ABCD$ is the required quadrilateral.

Calculation of Area:

Area of the quadrilateral $ABCD$ $= \frac{1}{2} \times d \times (h_1 + h_2)$ *sq. units*

$= \frac{1}{2} \times 6.4 \times (3.8 + 5.3)$

$= 3.2 \times 9.1 = 29.12\ cm^2$

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